

PRODUCT REQUIREMENT DOCUMENT (PRD)

Enterprise Strategic Initiative: Hybrid Monetization Stack Migration

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****Target Audience:**** Engineering Leads, Ad Ops Teams, Executive Stakeholders

****Status:**** Approved for Implementation

1. Executive Summary & Strategic Rationale

Our current digital media properties rely heavily on a legacy sequential waterfall setup within Google Ad Manager (GAM). This architecture limits our revenue potential by evaluating demand partners one after another, which inherently leaves premium programmatic dollars on the table. Furthermore, this method slows down page rendering times and creates major reporting discrepancies.

This document serves as the technical specification for migrating to a ****Hybrid Programmatic Architecture****. By integrating client-side Header Bidding (Prebid.js) with server-side exchange pathways (Google Open Bidding), we will maximize auction pressure on every single impression while protecting the user experience.

2. Technical Architecture & Bidder Mechanics

To optimize user page metrics while simultaneously scaling yield, our demand footprint will be split strategically across client-side and server-side environments.

A. Client-Side Prebid.js Configuration Rules

The engineering team must implement Prebid.js with a strict focus on latency management. Bids must be executed asynchronously to ensure content loads seamlessly.

* **Fail-Safe Timeout:** A hard timeout limit of **800ms** will be hardcoded. Any demand partner that fails to return a valid payload within this window is dropped from the auction.

* **Price Granularity:** Configured to `dense` (sliding scales tracking \\$.01 increments up to \\$5.00, and \\$.05 increments up to \\$10.00) to capture accurate yield data.

B. The Demand Matrix Setup

The following allocation rules dictate how our primary SSP partners are integrated into the new stack:

| Supply-Side Partner (SSP) | Integration Type | Operational Rationale & Performance History |

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| **Index Exchange** | Client-Side (Prebid.js) | High rich-media match rates; strictly requires client-side cookie syncing to capture premium demand. |

| **Magnite (Rubicon)** | Client-Side (Prebid.js) | Dominant partner for Private Marketplace (PMP) and Programmatic Guaranteed (PG) deals in North America. |

| **OpenX** | Server-Side (Open Bidding) | Provides consistent fill but introduces high client-side script overhead. Moved server-side to optimize DOM load times. |

| **PubMatic** | Server-Side (Open Bidding) | Excellent global scale. Using server-side routing minimizes cellular data drain and battery usage on mobile devices. |

3. Google Ad Manager (GAM) Targeting & Key-Value Mapping

To prevent line item inflation inside GAM while maintaining absolute precision for dynamic pricing strategies, we will utilize the following targeting configuration.

* **Targeting Key:** `hb\_pb` (Tracks the winning Prebid price bucket value)

* **Targeting Key:** `hb\_bidder` (Tracks the specific winning bidder name)

Line Item Traffic Mapping:

The ad server will read the custom keys passed from the page code via Google Publisher Tags (` googletag.cmd.push`) and map them immediately to custom Price Priority line items:

* When `hb\_pb = 0.10` , the ad server targets the line item valued at **\ \$0.10** (Priority 12)

* When `hb\_pb = 1.45` , the ad server targets the line item valued at **\ \$1.45** (Priority 12)

* When `hb\_pb = 10.00` , the ad server targets the line item valued at **Monetization Cap \ \$10.00**

4. Operational QA Checklist & Rollout Strategy

Before deploying the updated wrapper across 100% of our production web traffic, the AdOps team is required to execute a controlled canary test and run through the following operational verification process:

1. **Canary Allocation:** Route exactly 5% of live site traffic to the new hybrid setup for a 24-hour evaluation window.
2. **Console Telemetry Check:** Open browser developer tools on the staging environment and execute `pbjs.getBidResponses()` . Verify that all client-side partners are returning clean bid payloads without throwing syntax errors.
3. **Data Mapping Validation:** Ensure that computed `hb\_pb` and `hb\_bidder` strings are passing properly into the ad server call scripts.
4. **Discrepancy Baseline Guardrail:** Monitor the gap between **SSP Triggers** and **GAM Impressions Sold** . If this reporting discrepancy surpasses an 8% threshold during the testing period, the rollout will automatically pause and revert to the baseline waterfall to protect our revenue run-rate.